CSE451 Computer & Network Security

Assignment 1: Data Encryption Standard (DES)

It is required to implement the DES algorithm in C.

DES supplementary material:

https://en.wikipedia.org/wiki/DES_supplementary_material

Required build command:

```
gcc  -O3  studentID.c  -o  studentID
```

The submission is one C source file. No need to use any external libraries or headers.

Required program usage:

```
studentID.EXE  "e"  key  plaintext  ciphertext
studentID.EXE  "d"  key  ciphertext  plaintext
```

The command arguments (key, plaintext, and ciphertext) are all filenames. Use fopen(), fread(), fwrite(), and fclose() to access the data.

Use ECB block cipher mode: Divide the plaintext and ciphertext file into blocks and encode each block separately with the same key. Assume the message size is appropriate, and no padding is needed.

Declare the main function like this:

```c
int main(int argc, char **argv)
{
        return 0;
}
```

Then add your code.

Program speed is measured with the encryption and decryption of a large message. The provided message size is always a multiple of the block size, so do not implement any padding techniques. The fastest 3 submitted programs get a full mark in the next assignment. Copied sources get zero grade.